EXAMINATION: Magnetic Resonance Imaging (MRI) of the Brain with and without intravenous Gadolinium contrast.

CLINICAL INDICATION: A 64-year-old male presenting with new-onset generalized tonic-clonic seizures, progressive morning headaches, and cognitive slowing over three weeks.

TECHNIQUE: Multi-slice, multi-planar magnetic resonance imaging of the neurocranium was performed on a 3.0-Tesla scanner. Pulse sequences included axial T1-weighted, T2-weighted, 3D FLAIR, diffusion-weighted imaging (DWI) with ADC mapping, susceptibility-weighted imaging (SWI), and post-contrast 3D T1-weighted MPRAGE after intravenous administration of 10 mL of Gadavist.

FINDINGS:

Intra-Axial Mass Lesion:

There is an irregular, heterogeneously enhancing, necrotic mass lesion centered in the subcortical white matter of the right frontoparietal lobe measuring approximately 4.8 x 4.2 x 3.9 cm.

- T1/T2 Characteristics: The mass demonstrates a thick, irregular, nodular peripheral ring of T1-hypointense and T2-intermediate signal with a central area of fluid attenuation consistent with liquefactive necrosis.

- FLAIR: Significant surrounding hyperintense vasogenic edema is present, extending through the corona radiata into the right basal ganglia and corpus callosum splenium.

- SWI: Shows internal foci of magnetic susceptibility artifact consistent with microhemorrhages and tumor neovascularity.

- DWI/ADC: The central necrotic component shows facilitated diffusion, while the hypercellular enhancing peripheral margin demonstrates restricted diffusion with low ADC values (0.75 x 10^-3 mm2/s).

- Contrast Enhancement: Avid, irregular, rim-enhancement with nodular projections along the inner border is demonstrated.

Mass Effect and Herniation:

The mass causes substantial local mass effect with compression and effacement of the right lateral ventricle frontal horn and body. There is a 7 mm leftward midline shift of the septum pellucidum beneath the falx cerebri (subfalcine herniation). Early effacement of the right perimesencephalic cistern is present without frank uncal herniation.

Remainder of Neuroaxis:

No secondary satellite lesions identified. The major intracranial intracranial dural venous sinuses are patent.

IMPRESSION:

1. Large, necrotic, rim-enhancing right frontoparietal intra-axial mass lesion with extensive vasogenic edema and 7 mm subfalcine herniation, classic for high-grade diffuse astrocytic glioma (Glioblastoma, WHO Grade 4).

2. Emergent neurosurgical consultation recommended for decompressive resection.